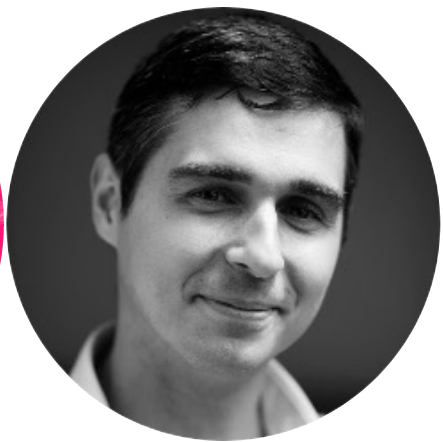


Mastering Looping

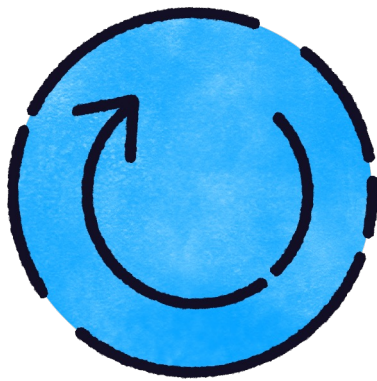


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Loops



```
while (daysLeft > 0) { }
```

```
do { } while (daysLeft > 0);
```

```
for(int i = 0; i < 10; i++) { }
```

```
for(String e : list) { }
```



```
int a = 1, b = 3;
```

```
int x = 0;
```

```
int y = 0;
```

```
while (a < 3) {
```

```
    x = a++;
```

```
    y = --b;
```

```
}
```

```
System.out.println(x);
```

```
System.out.println(y);
```



// a++ : use/print the var, THEN increment

```
int a = 5;
```

```
System.out.println(a++);
```

// 5

```
System.out.println(a);
```

// 6

// ++b : increment, THEN use/print the var

```
int b = 10;
```

```
System.out.println(++b);
```

// 11



Question

What will this print?

```
int x = 5;  
double y = 2 + 2 * x--;
```

10
decremented AFTER the operation

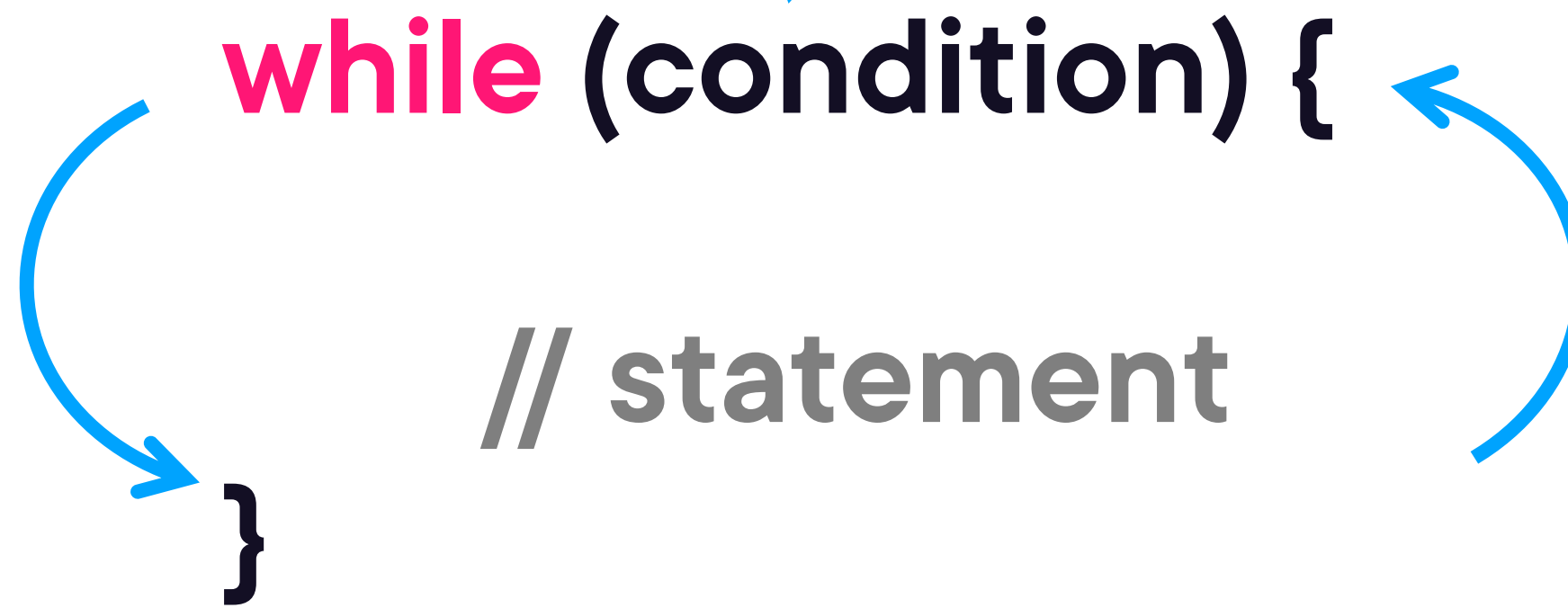
```
System.out.println(y);  
System.out.println(x);
```

12.0

4



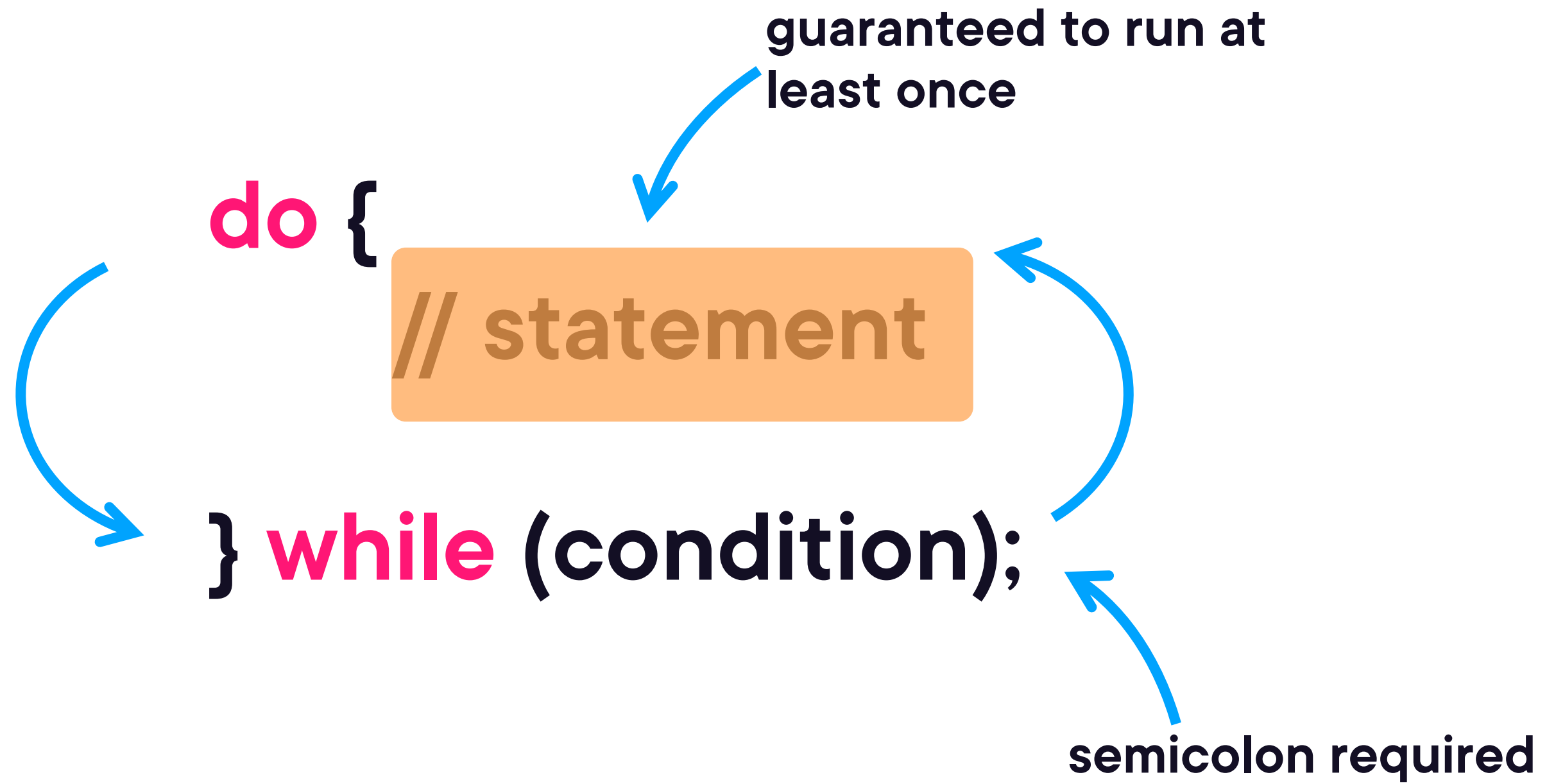
must be a boolean



while (condition) {
 // statement
}

The diagram illustrates the syntax of a while loop. The word **while** is in pink, and the rest of the code is in dark blue. A blue arrow points from the text "must be a boolean" to the **(condition)** part of the loop. Another blue arrow points from the opening curly brace **{** to the closing curly brace **}**, indicating the loop body. A third blue arrow points from the closing curly brace **}** back to the opening curly brace **{**, indicating the loop's repetition.





```
int daysLeft = 3;  
int daysWorked = 0;  
  
while (daysLeft > 0) {  
    System.out.println("work");  
    daysLeft--;  
    daysWorked++;  
}  
System.out.println(daysWorked);
```



```
int daysLeft = 3;  
int daysWorked = 0;
```

always false



```
while (daysLeft < 0) {
```

```
    System.out.println("work");
```

```
    daysLeft--;
```

```
    daysWorked++;
```

```
}
```

```
System.out.println(daysWorked);
```

never runs



```
int energyPointsLeft = 3;  
int hoursLeft = 3;  
int hoursWorked = 0;  
  
while (energyPointsLeft > 0 || hoursLeft >= 0) {  
    energyPointsLeft--;  
    hoursLeft--;  
    hoursWorked++;  
}  
System.out.println(hoursWorked);  
// 4
```



```
while(false) {  
    // body  
}
```



0 iterations

The diagram illustrates that a while loop with a false condition will not execute its body. An orange curved arrow starts from the opening curly brace of the loop and points back to the condition, indicating that the loop body is never reached.

```
do {  
    // body  
} while(false);
```



1 iteration

The diagram illustrates that a do-while loop with a false condition will execute its body exactly once before checking the condition. An orange curved arrow starts from the semicolon at the end of the loop and points back to the opening curly brace, indicating that the body is executed at least once.



```
for(initialization; booleanExpression; updateStatement) {  
    // body  
}
```

```
for( int i = 0 ; i < 10 ; i++ ) {  
    System.out.print(i + " ");  
}
```

0 1 2 3 4 5 6 7 8 9



Question

What will this print?

Can be anything:

$x = x * 2$

$x = x - 3$

...



```
for(int x = 10; x >= 8; --x) {  
    System.out.print(x + " ");  
}
```

10 9 8



init evaluate update

```
for( long x = 1, y = 13; x < 2 || y > 10; x++, y--) {  
    System.out.println(x + "-" + y);  
}
```

1-13

2-12

3-11



always false



```
for(int x = 10; x < 8; --x) {  
    System.out.print(x + " ");  
}
```



```
int i = 0; // fails too
```

```
for(int i = 0; i < 10; i++) {  
    System.out.print(i + " ");  
}
```

```
System.out.print(i); // does not compile
```



```
int i = 0;

for(i = 0; i < 10; i++) {
    System.out.print(i + " ");
}
System.out.print(i);
```



Different types not allowed



```
for(long x = 1, int y = 13; evaluate ; update) {  
  
}
```



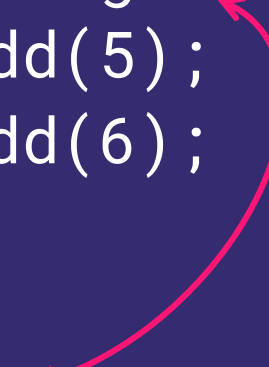
types must match

```
for (type e : collection) {  
    // statement  
}
```

array
List, Set, Queue
anything that
implements “Iterable”

```
String[] fruit = new String[2];  
fruit[0] = "apple";  
fruit[1] = "orange";  
  
for(String element : fruit) {  
    System.out.println(element);  
}
```

```
List<Integer> nums = new ArrayList<>();  
nums.add(5);  
nums.add(6);  
  
for(int element : nums) {  
    System.out.println(element);  
}
```





does not compile

```
String fruit = "apple";  
  
for(String e : fruit) {  
    System.out.println(e);  
}
```

```
List<Integer> nums = new ArrayList<>();  
nums.add(5);  
nums.add(6);
```



does not compile

```
for(String element : nums) {  
    System.out.println(element);  
}
```



**while, do/while, and classic
for loops may run infinitely**



Infinite Loop

```
int x = 0;  
while(x < 5) {  
    System.out.println("iteration: " + x);  
}
```

always true



Infinite Loop

all 3 statements are optional



```
for(;;);
```



Question 1

What will this print?

```
int x = 1;  
int y = 7;  
  
while (x > 0 && y <= 10) {  
    x--;  
    y++;  
}  
System.out.println(x + "-" + y);
```

0-8



Question 2

What is the result of this code?

```
int sum = 0;  
int a = 1;  
  
for(int a = 0, c = 1; a < 10; a = a + 2) {  
    sum = a + c;  
}  
  
System.out.println(sum);
```

Compilation error



Question 3

What is the result of this code?

```
int i;  
  
for(i = 2; i < 8; i *=2) {  
    i++;  
}  
  
System.out.println(i);
```

14

